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Fundamental and Molecular Mechanisms of Mutagenesis

Distinct difference in relative biological effectiveness of ^{252}Cf neutrons for the induction of mitotic crossing over and intragenic reversion of the *white-ivory* allele in *Drosophila melanogaster*

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Abstract

The relative biological effectiveness (RBE) of ^{252}Cf neutrons was determined for two different types of somatic mutations, i.e., loss of heterozygosity for wing-hair mutations and reversion of the mutant *white-ivory* eye-color, in *Drosophila melanogaster*. Loss of heterozygosity for wing-hair mutations results predominantly from mitotic crossing over induced in wing anlage cells of larvae, while the reverse mutation of eye-color is due to an intragenic structural change in the *white* locus on the X-chromosome. For a quantitative comparison of RBE values for these events, we have constructed a combined mutation assay system so that induced mutant wing-hair clones as well as revertant eye-color clones can be detected simultaneously in the same individuals. Larvae were irradiated at the age of 80 ± 4 h post-oviposition with ^{252}Cf neutrons or ^{137}Cs γ-rays, and male adult flies were examined under the microscope for the presence of the two types of clonal mosaic spots appearing. The induction of wing-hair spots per dose unit was much greater for ^{252}Cf neutrons than for ^{137}Cs γ-rays, whereas the frequencies of eye-color reversion were similar for neutrons and γ-rays. The estimated RBE values of neutrons were 8.5 and 1.2 for the induction of mutant wing-hair spots and revertant eye-color spots, respectively. These results indicate that the RBE of neutrons is much greater for mitotic crossing over in comparison to the intragenic *white-ivory* reversion events. Possible causes for the difference in RBE are discussed.

Keywords: Neutron mutagenesis; Relative biological effectiveness (RBE); Somatic mutation; *Drosophila melanogaster*

1. Introduction

The relative biological effectiveness (RBE) of neutrons compared with X- or γ-rays shows a large variation depending on various physical and biological factors [1–3]. Major biological factors which lead to different RBE values include differences in species or cell lines, developmental stages of organisms, cell cycle stages, and different biological endpoints such as lethality, mutation type and tumor induction. Thus, RBE is a complicated radiobiological parameter, and it is important to understand the underlying mechanisms that account for the variations in RBE.

RBE of neutrons was extensively investigated in the past with germ cell mutation systems in

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Drosophila [4–8]. These studies have shown that the effectiveness of neutrons to induce recessive lethal mutations was virtually the same for X- or γ-rays, whereas neutrons yielded high RBE values for chromosomal mutations such as translocations, minute deletions, rearrangements and dominant lethals.

In contrast to germ cell mutations, there are only a few studies on RBE for somatic mutation events in Drosophila [9,10]. Yet, mechanisms for induction of somatic types of mutation have drawn much attention in recent years because evidence is accumulating to show that the process of carcinogenesis is associated with the sequential occurrence of several somatic mutation steps involving various oncogenes and tumor suppressor genes [11].

In Drosophila, there are several somatic mutation assay systems such as the wing-spot test [12], the *white-ivory* reversion assay [13] and the *unstable-white-zeste* system [14]. The mutant wing spots mostly result from somatic recombination involving the two recessive genes, *multiple wing hairs* (*mwh*) and *flare* (*flr*) hair, both located on the third chromosome. In contrast, the reversion of the *white-ivory* (w^i) eye-color mutation is due to the excision of a duplicated 2.9-kb DNA fragment of the *white-ivory* gene [15]. By their high sensitivity to various mutagens, the two mutation assays seem superior to the conventional recessive lethal mutation test and are thus used in screening for environmental mutagens [13,16].

To clarify cellular and molecular mechanisms associated with the variation of RBE of neutrons, we compare the RBEs of ^{252}Cf neutrons for the induction of these two somatic mutation types. For this, a Drosophila strain was constructed which allows both types of mutations to be detected simultaneously in the same individuals. Here we report that RBE values of ^{252}Cf neutrons were distinctly different for somatic recombination and the intragenic reverse mutations.

2. Materials and methods

2.1. Strains and genetic markers

The *Drosophila melanogaster* strains, *y ; mwh jv* [12] and $flr^3/TM3$, *Ser* [16] used in the wing-spot assay were kindly provided by Dr. F.E. Würgler, University of Zürich, Switzerland. In the adult wing, flies homozygous for the recessive marker *mwh* produce 3 or more hairs on each wing epidermal cell, instead of a single smooth hair on each normal cell. The recessive marker *flr* is lethal in the homozygous state. However, when homozygous cells are produced somatically in heterozygous flies, they are viable as long as the clone size is small and produce a flare-shaped hair on each mutant cell. However, in the smallest clones the mutant phenotype is frequently not well expressed. The loci *mwh* and *flr* are located on the left arm of the third chromosome at map positions 0.3 and 38.8, respectively [17].

A strain with the genotype *Dp (1;1;1;1)* w^i [13] for the eye-color reversion assay was obtained from Dr. H. Ryo, Osaka University, Japan. This strain possesses 4 copies of the X-linked eye-color mutation *white-ivory* (w^i) in a tandem quadruplication (referred to as $(w^i)_4$ hereafter), and shows increased responsiveness to mutagens in comparison to the strain with a single w^i copy [13].

The strains *Basc* carrying multiple inversions on the X-chromosome and *C(1) RM, y*/Y with attached X-chromosomes are from our laboratory stock collection.

Detailed information on the genetic markers and their phenotypes is found in Lindsley and Zimm [17].

2.2. Construction of experimental stocks and culture of larvae

The third chromosome from the strain *y; mwh jv* was introduced into strains *C(1)RM, y*/Y and *Basc* to obtain strains with *C(1)RM, y/Y; mwh* females and *Basc; mwh* males, respectively. To construct the $(w^i)_4$ and *mwh* double mutant, $(w^i)_4$ males were crossed to *C(1)RM, y*/Y; *mwh* females and then the resulting $(w^i)_4$ male progeny flies were backcrossed to *C(1)RM, y*/Y; *mwh* females to produce $(w^i)_4$; *mwh* males.

For irradiation, male $(w^i)_4$ larvae trans-heterozygous for *mwh* and flr^3, $[(w^i)_4; mwh\ flr^+/mwh^-\ flr^3]$, were obtained from the following crosses: $(w^i)_4$; *mwh* males were crossed to *Basc; mwh* females. The female progeny $(w^i)_4$/*Basc; mwh* and males from stock $flr^3/TM3$ were kept separate as virgins for 3–5 days at 25°C. They were then mated at a ratio of 80 females to 40 males in 190 ml culture bottles containing 20 ml of Drosophila culture medium (150 g of dry yeast, 14 g of agar, 100 ml of molasses, and 5 ml of propionic acid in 1000 ml water) for about 12 h, and were then transferred to fresh culture bottles for egg laying during 8 h. After oviposition, parental flies were discarded and eggs were allowed to hatch at 25°C. For irradiation, the larvae were collected from the culture bottles 3 days later.

2.3. Radiation sources and dosimetry

A ^{252}Cf neutron source (71.4 GBq (1.93 Ci)) was purchased from Amersham International, Buckinghamshire, UK. Doses of ^{252}Cf neutrons and contaminating γ-rays were measured by paired ionization chambers, IC-17 and IC-17G (Far West Technology, Goleta, CA, USA), which were calibrated against the Japanese Association of Radiological Physicists (JARP) substandard ionization chamber. The dosimetry showed that 67% of the total absorbed dose was due to neutrons and 33% to γ-rays [18]. A ^{137}Cs γ-ray source (1.1 TBq (30 Ci)) was used as a standard radiation to determine the RBE of ^{252}Cf neutrons, and its γ-ray dose was measured by the JARP chamber within an error of 2–3%. These radiation sources are installed at the Research Institute for Nuclear Medicine and Biology, Hiroshima University.

2.4. Radiation exposure

Larvae were collected from culture bottles and placed in a plastic irradiation box made of tissue-equivalent material (Shonka A-150 TE), which has a 5 mm-thick front surface facing the radiation source and an inner volume of 3 mm in depth and 10 mm in diameter. The thickness of the front surface was adjusted to the wall thickness of the IC-17 ionization chamber. At 80 ± 4 h after oviposition, the larvae were exposed either to ^{252}Cf neutron and γ-ray mixed radiation or to ^{137}Cs γ-rays for 3 h. We used 3 dose levels for each radiation type, which were attained by changing the distance between the radiation source and the samples. Immediately after irradiation, all larvae were transferred to fresh culture bottles and incubated at 25°C.

2.5. Detection of mutations

Somatic reversions from the *white-ivory* to the wild-type phenotype were scored as eye spots composed of red colored ommatidia appearing on a *white-ivory* background in the eyes of adult males [19]. From the flies that emerged, $(w^i)_4$; *mwh* $flr^+/$ mwh^+ flr^3 males were collected and kept-on for 2 – 3 days so that we were able to identify the eye-color mosaic spots with certainty. The mosaic spots and their sizes (number of red ommatidia per spot) were monitored under a stereoscopic microscope at 40 × magnification. After inspection for the presence of red-eye spots, flies were stored in 70% ethanol.

To detect mutant wing spots, wings of the stored flies were surgically removed, washed with 99% ethanol, and then mounted on glass slides in Faure's solution [12]. Since the frequency of induced wing spots is known to be considerably high, subsamples of the stored flies from the $(w^i)_4$ assay were used for wing analysis. Scoring for the presence of mutant wing spots and spot size determination (number of mutant cells per spot) were carried out under the microscope at 400 × magnification. Three types of mosaic spots are recovered with this system: (1) *mwh/flr* twin spots composed of a *mwh* area and an adjacent *flr* area; (2) *mwh* single spots; and (3) *flr* single spots. We have omitted the *flr* single spots from the present analysis, because they constitute only a small fraction of the total of mutant spots, and are considered to originate from mechanisms other than mitotic crossing over [10,20].

3. Results

3.1. Induction of wing mosaic spots

A total of 2250 wings from $[(w^i)_4$; *mwh* $flr^+/$ mwh^+ $flr^3]$ males irradiated either with ^{252}Cf radiation or ^{137}Cs γ-rays were analyzed for the induction of *mwh* single spots, *mwh/flr* twin spots, whereby the size (number of mutant cells) of each spot was also determined. As recommended by Graf et al. [12], *mwh* single spots were classified as large spots composed of 3 or more mutant cells or small spots of one or two cells. We considered large *mwh* spots and

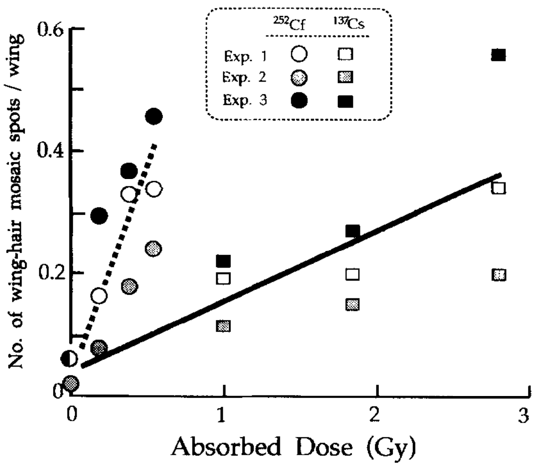

Fig. 1. Fig. 1Induction of wing-hair mosaic spots by mixed neutron/γ-radiation of ^{252}Cf or ^{137}Cs γ-rays. Larvae were irradiated either with ^{252}Cf radiation or ^{137}Cs γ-rays at the age of 80 ± 4 h post-oviposition, and the presence of wing-hair mosaic spots on the wings of adult flies was examined under the microscope. The ordinate shows the sum of the frequencies of large *mwh* single and *mwh/flr* twin spots, the large majority of which originate from somatic crossing over induced in wing anlage cells of the larvae. The results of 3 independent experiments are shown. The straight and dotted lines represent linear regressions fitted to the observation data.

mwh/flr twin spots as originating mainly from somatic crossing over.

Fig. 1 shows the frequencies of somatic crossing over observed in 3 independent experiments, plotted as a function of dose for both types of radiation. The frequencies correspond to the added numbers of large *mwh* spots plus *mwh/flr* twin spots divided by the number of wings examined. The frequencies of mosaic spots at each dose varied considerably among experiments, but within experiments they appeared to increase fairly linearly with dose of either ^{252}Cf mixed radiation or ^{137}Cs γ-rays. In any case, it is clear from the graph that ^{252}Cf radiation was much more effective than ^{137}Cs γ-rays for the induction of wing-hair somatic mutations. Numerical values obtained from individual experiments were pooled and summarized in Table 1.

To determine the induction rate, a linear model was fitted to the dose–response data for each radiation, and the regression coefficients were estimated by the least-square method. The rate of γ-ray induced events was $(1.09 \pm 0.11) \times 10^{-3}$/wing/cGy for irradiation of larvae 80-h after oviposition. This is in good agreement with rates reported previously: $(1.62 \pm 0.09) \times 10^{-3}$ obtained following irradiation of 75-h-old larvae with X-rays [21] and $(1.74 \pm 0.22) \times 10^{-3}$ for 72-h-old larvae irradiated with ^{60}Co γ-rays [9]. On the other hand, the induction rate by ^{252}Cf mixed radiation was $(6.56 \pm 0.60) \times 10^{-3}$/wing/cGy, which is similar to our previous determination of $(9.83 \pm 0.15) \times 10^{-3}$ [9]. Apparently, ^{252}Cf radiation was about 6 times more effective than ^{137}Cs γ-rays for the induction of wing mosaic spots resulting mainly from crossing over events. From these results, the RBE of ^{252}Cf neutrons was calculated as described below.

Table 1
Frequencies of mutant wing spots induced at larval stage by irradiation with ^{252}Cf radiation or ^{137}Cs γ-rays

Radiation source	Dose (cGy)	No. of wings observed [a]	No. of mutant spots (frequency) [b]			Frequency of somatic crossing over [e] (% ± S.E.)
			Small *mwh* single [c] (%)	Large *mwh* single [d] (%)	*mwh/flr* twin(%)	
Control	0	450	55 (12.2)	10 (2.2)	7 (1.6)	3.8 ± 0.9
^{252}Cf radiation	18.1	450	85 (18.9)	60 (13.3)	21 (4.7)	18.0 ± 2.0
	37.6	300	32 (10.7)	67 (22.3)	21 (7.0)	29.3 ± 3.1
	54.1	150	27 (18.0)	39 (26.0)	13 (8.7)	34.7 ± 4.8
^{137}Cs γ-rays	99	450	78 (17.3)	56 (12.4)	23 (5.1)	17.6 ± 2.0
	184	300	43 (14.3)	45 (15.0)	17 (5.7)	20.7 ± 2.6
	279	150	26 (17.3)	37 (24.7)	18 (12.0)	36.7 ± 4.9

[a] Pooled data from 3 independent experiments (cf. Fig. 1).
[b] Frequency of mutant spots per 100 wings.
[c] Mutant single spots composed of one or two cells.
[d] Spots composed of 3 or more cells.
[e] Sum of the frequencies of large *mwh* single and *mwh/flr* twin spots.

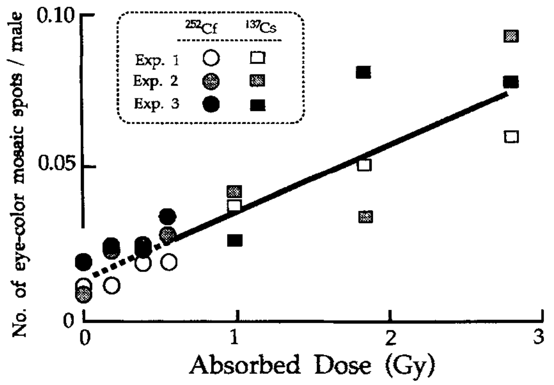

Fig. 2. Frequency of w^i reverse mutations induced by ^{252}Cf radiation or ^{137}Cs γ-rays. Eye-color mosaic-spot induction was determined on the same samples of adult flies used for wing-spot analysis.

3.2. Induction of eye-color mosaic spots

Somatic reversions of the *white-ivory* gene were detected as mosaic spots composed of wild-type and red-colored eye facets in flies exposed to either ^{252}Cf radiation or ^{137}Cs γ-rays at the larval stage. Frequencies of eye-color mosaic spots per male fly were determined in 3 independent experiments, and the results are plotted against radiation dose in Fig. 2. The pooled data are summarized in Table 2. There was quite some variation in the mutant frequencies among the triplicate experiments at each exposure dose. However, the average frequency derived from the pooled data increased almost linearly with dose for either radiation. The rates of induction of mosaic spots estimated by linear regression analysis were $(2.22 \pm 0.38) \times 10^{-4}$ and $(2.48 \pm 1.53) \times 10^{-4}$/male/cGy for ^{137}Cs γ-rays and ^{252}Cf radiation, respectively. As expected, the observed rate of induction by γ-rays was about 3-fold greater than that induced by X-rays in the ordinary single *white-ivory* (w^i) system [22], because of the increased number of target genes present in the $(w^i)_4$ system. The data presented in Fig. 2 for $(w^i)_4$ males indicate that the mutagenic efficiency of ^{252}Cf radiation was almost equal to that of ^{137}Cs γ-rays for the induction of eye-color mosaic spots, which is in marked contrast to the case of wing-spot induction.

Table 2
Frequencies of somatic reversion of the w^i eye-color mutation induced by ^{252}Cf radiation or ^{137}Cs γ-rays

Radiation source	Dose (cGy)	No. of males observed [a]	No. of mutant spots [a]	Frequency (% ± S.E.)
Control	0	881	11	1.2 ± 0.4
^{252}Cf radiation	18.1	844	16	1.9 ± 0.5
	37.6	561	12	2.1 ± 0.6
	54.1	232	6	2.6 ± 1.1
^{137}Cs γ-rays	99	878	32	3.6 ± 0.6
	184	643	32	5.0 ± 0.9
	279	332	26	7.8 ± 1.5

[a] Results of 3 independent experiments are pooled (cf. Fig. 2).

3.3. Relative biological effectiveness (RBE) of ^{252}Cf neutrons

^{252}Cf radiation consists of mixed neutron and γ-ray radiation; 67% of a total tissue dose is contributed by neutrons and the remaining 33% by γ-rays [18]. Hence, the mutation rate μ_n due to the presumed ^{252}Cf neutrons alone can be calculated according to Ayaki et al. [10]:

$$\mu_n = \frac{3}{2}\mu_{n+\gamma} - \frac{1}{2}\mu_\gamma$$

where $\mu_{n+\gamma}$ is the rate of mutation induced by the mixed neutron and γ-ray radiation of ^{252}Cf, and μ_γ is the rate induced by γ-rays. On the assumption that μ_γ is equal for ^{137}Cs γ-rays and the contaminating ^{252}Cf γ-rays, the ratio μ_n/μ_γ gives the RBE of ^{252}Cf neutrons.

On account of the presently observed rates of mutation induction, the derived RBE of ^{252}Cf neutrons was 8.5 for the induction of wing spots, whereas it was 1.2 for the induction of eye spots.

4. Discussion

In the present study, we have analyzed the RBE values of ^{252}Cf neutrons for induction of two different types of somatic mutation events, i.e., interchromosomal mitotic crossing over as assessed in the wing-spot test [12] and an intragenic specific type of reversion as assessed in the $(w^i)_4$ eye-spot test [13]. The observed RBE was 8.5 for induction of mitotic crossing over, thus confirming the large RBE values previously reported, i.e., 8.0 for the induction of large *mwh* single and *mwh/flr* twin spots by ^{252}Cf neutrons [9], and 6.9 for large *mwh* single spots by fission neutrons [10]. By contrast, the RBE for the eye-color reverse mutations presently determined was 1.2, which means that ^{252}Cf neutrons were no more effective than γ-rays for the induction of this type of mutation. Although a rather small dose range of ^{252}Cf radiation was used in this study (no greater than 54 cGy), we think that the observed linear dose–response curve can be extended to much higher dose range for the following reasons. We have recently analyzed the induction of eye-color mosaic spots in the $(w^i)_4$ system by accelerated carbon ions with average linear energy transfer (LET) of 62 keV/μm, which is similar to the LET of ^{252}Cf neutrons (40 keV/μm). We have found that the w^i reversion frequency increased linearly with the dose of carbon ions over a dose range from 50 to 600 cGy, with reverse mutation rates of 2.6 and 2.4×10^{-4}/male/cGy by carbon ions and X-rays, respectively (Yoshikawa et al., to be published elsewhere). These values are almost equal to the present rate of 2.61×10^{-4}/male/cGy for ^{252}Cf neutrons, showing that the induced frequencies of w^i reversion are very much the same for different qualities of radiation, such as ^{252}Cf neutrons, accelerated carbon ions, γ-rays and X-rays. Hence, it can be reliably concluded that the RBE of ^{252}Cf neutrons is very close to one.

This study employed a new test system by which the two types of somatic mutation can be monitored in the same flies. This eliminates possible experimental biases in comparing RBE values for the two different mutational events, such as may arise from dosimetry problems and variations in the physiological conditions. Thus, the observed distinct difference in the RBEs for interchromosomal mitotic crossing over and intragenic reversion events suggests that the primary DNA lesions and/or the repair processes responsible for these two genetic endpoints are quite different.

It is generally accepted that all *mwh/flr* twin spots and most of the large *mwh* single spots result from somatic recombination, i.e., mitotic crossing over [10,12,20]. With regard to the induction mechanism for mitotic crossing over, Kondo and collaborators [10,21,23] proposed that: (1) radiation-induced DNA double strand breaks (DSBs) are repaired by a postreplication repair mechanism; and (2) unrepaired DSBs carried over into the G_2 phase are dealt with by recombinational repair. Thus, large *mwh* single and *mwh/flr* twin spots would result from G_2-specific recombinational repair of potentially lethal DSBs. This hypothesis explains well the large RBE values of high LET radiation, e.g. neutrons, as DSBs induced by neutrons are more abundant and more densely distributed than those induced by X- or γ-rays [2,3,24]. They may also stay open for a longer time: at least it was observed that a fraction of irreparable DSBs increased sharply in abundance with increasing LET of radiation, at least up to 200 keV/μm [25]. The frequency of large *mwh* spots induced by X-rays was several times higher in the postreplication repair-deficient strain *mei-41*, compared to wild type [21]. This finding is in line with the idea that mitotic crossing over results from recombinational repair of long-persisting DSBs.

The *white-ivory* eye-color mutation is caused by a tandem duplication of a 2.9-kb DNA sequence in the *white* locus, and reversion to wild type is associated with the excision of one of the copies of the duplicated sequence [13,15]. Ryo and Kondo [22] reported that X-rays were about two orders of magnitude more effective than ultraviolet light (UV) for the induction of reverse mutation in relation to representative DNA damage produced by each radiation. They suggested that the excision of one 2.9-kb copy occurs efficiently only at DNA strand breaks, the major damage produced by X-rays. The frequency of reversions was not affected by the dose rate of X-rays, indicating that the reverse mutations probably result from a single ionization event [19]. This and the observed small RBE of ^{252}Cf neutrons suggest that single strand breaks (SSBs) may be the primary DNA lesions causing the w^i reverse mutations. The following observations support this idea: (1) the yields of SSBs induced by X- or γ-rays in mammalian cells are similar or slightly larger than those induced by high LET radiation, including neutrons [2,26]; (2) X-ray induced SSBs are rejoined rapidly with a half-life of about 5 min, as the repair rate is not affected by dose rate or split-dose, while split-dose irradiation results in a marked reduction of persisting DSBs [27]; (3) after treatment with X-rays and UV, frequencies of w^i reversion were significantly lower in the postreplication repair deficient strain *mus-104*, compared with repair-proficient strains [22]; and (4) in *Escherichia coli*, strains defective in postreplication repair were unable to repair X-ray-induced SSBs [28,29], and were not mutable by X-rays [30].

We assume that most SSBs produced in the duplicated 2.9-kb fragment are probably repaired, but a small fraction of the breaks may lead to the excision of the duplicated copy, possibly through the pairing of the duplicated sequence followed by subsequent intrachromosomal recombination [19,22]. The latter process of homologous pairing and recombination could be, at least in part, operated by a postreplication repair system.

Our present study revealed distinctly different RBE values of ^{252}Cf neutrons for the induction of mitotic crossing over and w^i reverse mutations. On account of the small RBE (1.2) for the induction of w^i reverse mutation, we propose that DNA SSB may be the primary cause of this type of mutation, in contrast to mitotic crossing over which is thought to originate from long-persisting DSBs. We cannot completely rule out the possibility that the w^i reverse mutations could be caused by sparsely formed DSBs produced by single ionization events [31], and that mitotic crossing over could result from clustered DSBs. However, this seems unlikely in view of the greater yield of DSBs by neutrons compared with γ-rays and the repair characteristics of DSBs. In any case, more studies are necessary to elucidate the different molecular mechanisms leading to these interchromosomal and intragenic types of mutation. We believe that our simultaneous wing and eye spot test system in Drosophila will be very useful for this purpose.

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